

3 The photograph shows one species of the genus *Cabomba*, an aquatic plant.



(Source: © Dorling Kindersley Ltd/Alamy Stock Photo)

Cabomba grows in streams and ponds in many parts of the world.

It adds oxygen to the water.

A student observed that there were fewer *Cabomba* plants growing in the shaded parts of a pond.

The student formed the following hypothesis.

The greater the light intensity the faster the rate of photosynthesis in *Cabomba* plants.

Plan an investigation to test this hypothesis.

(a) State **one** safety issue you would need to take into account.

(1)

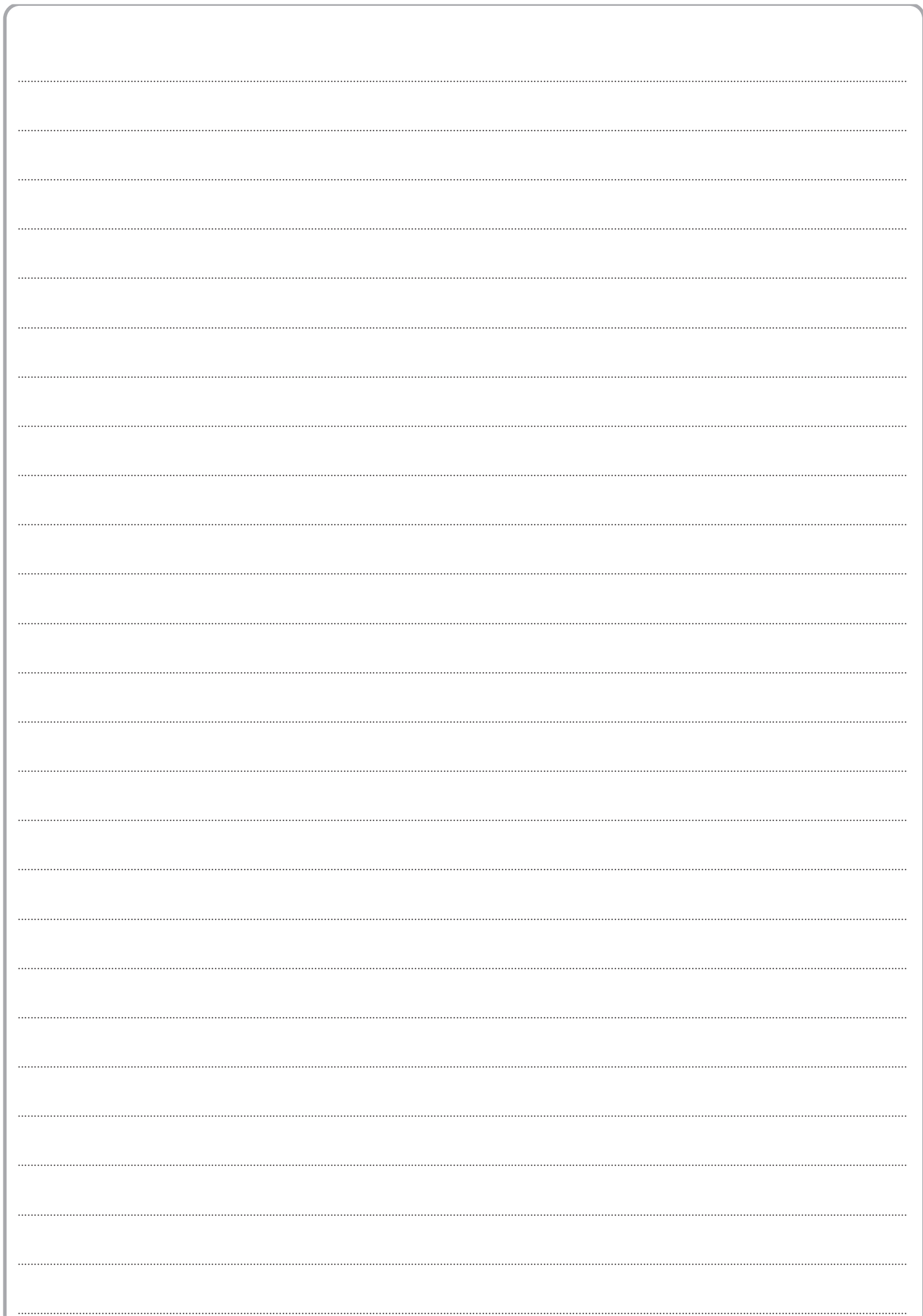
(b) Describe the preliminary practical work that you might undertake to ensure your proposed method would provide meaningful data.

(3)

(c) Devise a detailed method, including an explanation of how you would control and monitor important variables.

(10)

This image shows a single sheet of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting or typing. There are no margins, text, or other markings on the paper.



(d) Describe how your results should be recorded, presented and analysed in order to draw conclusions from your investigation.

(4)

[illegible]

(e) Suggest **three** limitations of your proposed method.

(3)

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2.....

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3.....

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(Total for Question 3 = 21 marks)

TOTAL FOR PAPER = 50 MARKS